

Tutorial Serie 1

Objectives : to master the basic instructions (assignment, reading, writing)

Exercise 1

Let the following problems:

1. Calculation of the sum of two complex numbers.
2. Calculation of the product of two complex numbers.
3. Determination of the type of an angle θ expressed in degrees. Knowing that an angle can be :
 - Zero (or Null) ($\theta = 0^\circ$),
 - Acute ($0^\circ < \theta < 90^\circ$),
 - Right ($\theta = 90^\circ$),
 - Obtuse ($90^\circ < \theta < 180^\circ$),
 - Straight ($\theta = 180^\circ$),
 - Reflex ($180^\circ < \theta < 360^\circ$),
 - Complete (or Full) ($\theta = 360^\circ$).
4. Deliberation of a group of 20 students by displaying for each student: his name, his first name, his identification number, his average and the decision of the jury of deliberation «Admitted» or «Postponed ». Knowing that:
 - All students in the group follow 4 modules (M1, M2 M3 and M4).
 - The respective coefficients of modules are: 2, 1, 3 and 2.
 - The student is declared «Admitted» if his average is greater than or equal to 10 otherwise is declared «Postponed».

Question:

Identify for each of the above issues:

- The input data,
- The output results
- The main stages (steps) of resolution.

Exercise 2

1. Knowing that $a = 4$, $b = 5$, $c = -1$ and $d = 0$, evaluate the following logical expressions:
 - $a < b$ **AND** $c \geq d$
 - **NOT** ($a < b$) **OR** $c \neq b$
 - **NOT** ($a \neq b^2$) **AND** $a * c < d$
2. Knowing that: $A = \text{TRUE}$, $B = \text{FALSE}$, $C = \text{TRUE}$; evaluate the following logical expressions:
 - A **OR** B **AND** A **OR** C
 - **NOT** (A) **AND** B **OR** A **AND** **NOT** (B)
 - A **AND** B **AND** B **AND** C **OR** C **AND** A
 - A **AND** B **OR** B **AND** C **ET** C **AND** A
 - A **OR** A **AND** B **AND** A **OR** B **AND** C
3. Knowing that $A=3$, $B=4$, $X=3.0$, $Y = -1.0$, $C='K'$ and $F=\text{False}$.
Indicate the evaluation order and the value of each of the following expressions:
 - $-X * A + Y$,
 - $B - A / Y + 2$,
 - $4 + A * 4 - B + A + 2 \ 4 / Y - 9 + 6 * 3$,

- $B - 2/Y/3/2 * 5 * X - 4/2$,
- $F \text{ AND NOT } (C < 'A')$,
- $X \leq Y \text{ OR } A > B$,

Exercise 3

These are the following algorithms:

```

Algorithm Algo_01;
Var A, B, C: integer ;
      D: Boolean;
Begin
  A  $\leftarrow$  5;
  B  $\leftarrow$  6 ;
  C  $\leftarrow$  A + B*2 + 3;
  D  $\leftarrow$  (C mod A) < (C div B) ;
  Write (A,B,C,D);
End.

```

```

Algorithm Algo_02;
Var A, B, C: integer ;
      D: Boolean;
Begin
  Read (A);
  Read (B);
  C  $\leftarrow$  A + B*2 + 3;
  D  $\leftarrow$  (C mod A) < (C div B) ;
  Write (A,B,C,D);
End.

```

```

Algorithm Algo_03;
Var x, y: real;
Begin
  x  $\leftarrow$  10;
  y  $\leftarrow$  x * 2;
  Write (x, " *2= ", y);
End.

```

```

Algorithm Algo_04;
Var x, y: real;
Begin
  Read (s);
  y  $\leftarrow$  x * 2;
  Write (x, " *2= ", y);
End.

```

Questions:

1. Perform the trace of the algorithms **Algo_01** and **Algo_02**.
2. What results does the Algo_03 algorithm produce.
3. What results does the Algo_04 algorithm produce.
4. What is the relationship between the last two algorithms.

Exercise 4

These are the following expressions "Exp1, Exp2 and Exp3":

$$\text{Exp1} = \frac{5(a - cd)^2 - 3bc}{2be + ac}$$

$$\text{Exp2} = \frac{-a + 3 - 5 * \frac{b}{c}}{\frac{2c}{d} - \frac{2c - 3}{b}}$$

$$\text{Exp3} = a + b - c - \frac{a + c * d - b + 5}{d + \frac{c}{2}}$$

Without mathematical simplifications, write instructions equivalent to the previous expressions.

Exercise 5

Write **an algorithm and its C program** that allows to calculate and display the square and the double of an integer introduced by the user.

Exercise 6

Writing an **algorithm** and its **C program** qui can read three (3) real numbers and calculate and display their sum, their product and their average.

Exercise 7 -optional-

Let the following algorithm :

```
Algorithm Algo_05 ;  
Var  A, B : entier ;  
Begin  
    Read (A);  
    Read (B);  
    A  $\leftarrow$  B;  
    B  $\leftarrow$  A;  
    Write (A,B);  
End.
```

Questions:

1. Perform the trace of the Algo_05 **algorithm** by entering 5 and 10 as values of A and B.
2. Can the algorithm instructions exchange the two values of B and A?
3. Reverse the order of the instructions (A $\leftarrow$ B;) and (B $\leftarrow$ A;) and perform the trace of the new algorithm.
4. If the value exchange problem of A and B is not resolved, propose an algorithm to solve this problem.

Exercise 8 -optional-

Let A, B and C be three points in the plane defined by their coordinates $A(x_A, y_A)$, $B(x_B, y_B)$ and $C(x_C, y_C)$.

Write an algorithm that determines and displays the nature of the ABC triangle.

(An ABC triangle is either isosceles, equilateral or any.)